Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – PERIODIC TEST 09
Chapter 1: Real Numbers

Time: 60 Minutes
MM: 30

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based question of 4 marks with internal subparts.
7. Use of calculators is strictly prohibited. All mathematical steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. If $a = 2^3 \times 3^2$ and $b = 2^2 \times 3^3$, then the value of $\frac{\text{LCM}(a,b)}{\text{HCF}(a,b)}$ is: [1]

(a) 6 (b) 12 (c) 36 (d) 18

Q2. If $x = 2^3 \times 3 \times 5^2$ and $y = 2^2 \times 3^3$, then HCF(x, y) is equal to: [1]

(a) 12 (b) 24 (c) 36 (d) 48

Q3. The number of consecutive trailing zeros in the number $N = 2^3 \times 3^4 \times 5^4 \times 7$ is: [1]

(a) 3 (b) 4 (c) 7 (d) 12

Q4. The exponent of 2 in the prime factorisation of 144 is: [1]

(a) 2 (b) 3 (c) 4 (d) 6

Q5. If p and q are positive integers such that $p = ab^3$ and $q = a^4b$, where a and b are prime numbers, then HCF(p, q) is: [1]

(a) ab (b) a^2b^2 (c) a^4b^3 (d) a^3b^2

Q6. **Assertion (A):** The HCF of two consecutive even natural numbers is 2.
Reason (R): Every even natural number is completely divisible by 2. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. Prove that $3 + 2\sqrt{5}$ is an irrational number, given that $\sqrt{5}$ is irrational. [2]

Q8. Prove that $5 - \sqrt{3}$ is an irrational number, given that $\sqrt{3}$ is irrational. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. In a school library, there are 780 books of English literature and 364 books of Hindi literature. The librarian wants to arrange them in shelves such that each shelf has the same number of books of the same language. Find the maximum number of books per shelf and the total number of shelves required. [3]

Q10. Prove that if p and q are distinct prime numbers, then $\sqrt{p} + \sqrt{q}$ is an irrational number. [3]

SECTION D: Long Answer Type (2 × 5 = 10 Marks)

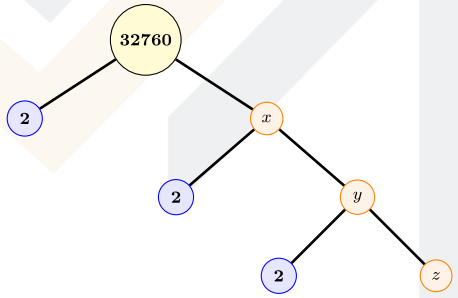
Q11. A dairy distributes milk from three large containers holding 403 litres, 434 litres, and 465 litres of milk respectively. (i) Find the maximum capacity of a measuring container that can measure the milk of all three containers in an exact number of times. (ii) How many times must this measuring container be used to empty each container? (iii) Find the total number of times the measuring container is filled. [5]

Q12. A warehouse floor of length 15 m 17 cm and breadth 9 m 2 cm is to be paved with the minimum number of identical square tiles. (i) Find the side length of the largest square tile that can be used. (ii) Calculate the exact number of tiles required. (iii) If each tile costs Rs 45, compute the total expenditure for paving the floor. [5]

SECTION E: Case Study Based Question (1 × 4 = 4 Marks)

Q13. CASE STUDY: CASE STUDY 09: MATHEMATICAL FACTOR TREE PRIME DECOMPOSITION [4 Marks]

During a mathematics laboratory session on the Fundamental Theorem of Arithmetic, a student investigates the prime factorization of large composite integers by constructing recursive factor trees. For the number 32760, the student creates a tree where the number branches into 2 and a composite node x; node x branches into 2 and node y; node y branches into 2 and node z; and node z branches into 3 and a node $w = 1365$.



(i) Determine the numerical values of the missing nodes x, y, and z. [2]

(ii) Write the complete prime factorisation of 32760 in canonical exponential form. [1]

(iii) State the Fundamental Theorem of Arithmetic. [1]

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